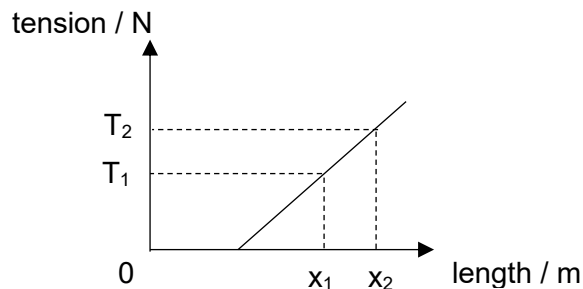


- 7 The graph below shows how the length of a wire changes with the tension in the wire.



What is the extra energy stored in the wire when the tension is increased from T_1 to T_2 ?

A $\frac{1}{2} (T_2 + T_1)(x_2 - x_1)$

B $\frac{1}{2} (T_2 + T_1)(x_2 + x_1)$

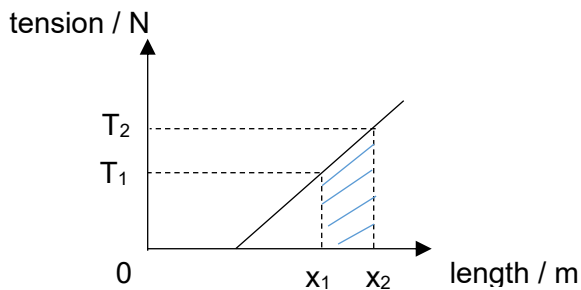
C $\frac{1}{4} (T_2 + T_1)(x_2 - x_1)$

D $\frac{1}{4} (T_2 + T_1)(x_2 + x_1)$

Ans: A

Extra energy stored is the shaded trapezium.

Area of dotted trapezium is $\frac{1}{2} (T_2 + T_1)(x_2 - x_1)$



8

In the diagram below, a body S of weight W hangs vertically by a thread tied at O to the string.